Project Documentation: Student Performance Analytics System

1. Project Title

Student Performance Analytics System

Student Name: [Enter your full name]

College: [Enter your college name]

Internship: Python with AI

Repository Link: [Paste your GitHub repository URL after uploading]

2. Objective

The objective is to create a Python program that reads student marks and attendance from a CSV file, analyzes the information, and presents useful summaries. It helps organize student results and quickly identify strong and weak subject performance.

3. Technologies Used

• Python: program logic, conditions, loops, and functions.

• Pandas: reads the CSV file and organizes data in a DataFrame.

• NumPy: calculates totals and averages.

• CSV: stores the sample student dataset.

4. Dataset Description

The sample dataset contains 20 fictional student records. Columns include Student ID, Name, Department, marks in Math, Physics, Chemistry, English and Computer Science, and Attendance Percentage. Each subject is marked out of 100. Attendance is represented as a percentage.

5. Implementation

1. `pandas.read_csv()` loads `students.csv` into a DataFrame.

2. The program checks that the required columns exist.

3. NumPy calculates each student's total and average marks.

4. The `assign_grade()` function assigns a grade from the student's average.

5. The `check_pass_fail()` function uses a loop and condition to check whether every subject mark is at least 35.

6. Pandas calculates the class summary and subject-wise averages.

7. The program sorts students by average marks and displays the top three.

8. The calculated table is saved to `student_analysis_results.csv`.

6. Key Features

• Student-wise total and average marks.

• Grade assignment.

• Pass/fail status.

• Class average, highest average and lowest average.

• Number of passed and failed students.

• Average attendance.

• Subject-wise averages.

• Top three performers.

• Export of analysis results to CSV.

7. Output / Screenshots

After running `python main.py`, add screenshots of:

• The student-wise results table.

• Class summary.

• Subject-wise averages.

• Top three performers.

• The generated `student_analysis_results.csv` file, if useful.

Only add screenshots from your actual run.

8. Final Outcome

The program reads the sample dataset, processes the marks, assigns grades, identifies pass/fail status, summarizes class and subject performance, and saves the results to a new CSV file. It demonstrates basic Python programming and data analysis using NumPy and Pandas.

9. Challenges & Learning

Complete this section in your own words after testing the project. Possible learning points, if they match your experience:

• Learning how to load a CSV file using Pandas.

• Understanding how a DataFrame stores tabular data.

• Using NumPy to calculate totals and averages.

• Applying loops, conditions and functions to student records.

• Checking and fixing file-path or library-installation errors.

10. GitHub Repository

Paste the URL of your completed repository here: [GitHub URL]

Grading Rules

• A+: average >= 90

• A: average >= 80 and < 90

• B: average >= 70 and < 80

• C: average >= 60 and < 70

• D: average >= 50 and < 60

• F: average < 50

A student passes only if every subject mark is at least 35. Grade and pass/fail are separate measures.